How to Not Get Run Over By Robots!

Podcast

Summary

During this special live episode of the *Approximately Correct* podcast, recorded at UpperBound Live, hosts chat with Mo Chen, a Professor in the School of Computing Science at Simon Fraser University (SFU) and Canada CIFAR AI Chair at Amii. Chen, who directs the Multi-Agent Robotic Systems Lab at SFU, specializes in developing algorithms that help robots interact safely and naturally with humans. In this conversation, Chen shares how his early experiences in robotics, building metal-bodied machines as a student and advancing research in robotic safety, shaped his mission to ensure robots can navigate the real world. Chen provides insights on the future of human-robot collaboration, from autonomous luggage and hospital assistance robots, to his ongoing work designing systems that move with human intent.

Robots in human spaces face two big hurdles: unspoken rules and scarce data. Unlike roads, with lanes, lights, and signage, indoor environments run on situational norms that are hard to formalize and vary by crowd density. Chen contrasts this with self-driving vehicles, where vast amounts of road data have been collected for years. In indoor or human-centered spaces, however, gathering that level of data would require constant camera surveillance over long periods, something society may not want. As a result, robots have historically learned to operate with limited data. Chen explains that robotics research has been able to progress even without modern data-driven AI because there is "classical" AI, which focuses on search, planning, and pathfinding. The challenge arises in dynamic, uncontrolled environments, where tuning parameters manually becomes impractical, and a single configuration cannot handle every changing situation. For these complex and intuitive contexts, Chen argues that the solution is to combine the strengths of classical AI with modern, data-driven AI.

Computer vision is another essential branch of AI for robotics. It allows robots to reliably tell a person from a dog, or a person pushing a cart, each of which moves differently and requires distinct avoidance strategies. Classical AI methods also remain valuable; for instance, a Bayesian approach helps predict the likelihood or probability of certain situations. While humans have built-in common sense, Chen notes that robots do not. He sees promise in foundational models, which capture elements of human common sense, but emphasizes that researchers must learn how to distill these models into robot behavior and integrate them with classical techniques.

In practice, when today's service robots encounter uncertainty, their default response is to "stop when in doubt."While this minimizes collisions, it can create new problems. For example delivery robots have clogged sidewalks in major cities, or it is possible for them to form immobile queues. Many near-term deployments still rely on human teleoperators. Chen suggests a pragmatic metric for progress is improving the

human-to-robot supervision ratio over time; this is much like Vancouver's SkyTrain control room, where only a few staff oversee dozens of autonomous trains.

One project Chen's team is developing to test robots in uncontrolled environments is the "follow-ahead" project, inspired by autonomous suitcases. In this case, the robot stays ahead of a person, a design useful in airports or hospitals. The system must predict human motion, navigate safely, and account for how the person reacts to the robot's movements. Chen's group experiments with several approaches, using reinforcement learning to teach robots to occupy "good spots," analyzing human skeleton motion to predict where someone will be in two seconds, and employing tree search methods inspired by AlphaGo to plan multi-step interactions.

The podcast emphasizes that capability is not the most important factor in robotics. Robots should respect boundaries (for example, not retrieving a locked diary), keep humans in the loop for decisions that require human judgment, and be transparent about their actions. Above all, robots should be designed to operate safely and recognize when they are uncertain or lack knowledge. Technical progress must always be paired with ethics and user education.

Key Terms

Term	Definition
Bayesian Approach	A statistical method for estimating the probability of events or conditions based on prior knowledge and new evidence.
Computer Vision	A field of AI that enables machines to interpret and understand visual information (e.g., identifying objects, people, or movement).
Foundational Models	Large AI models (like GPT or CLIP) trained on massive datasets that can be adapted to many different tasks, often capturing aspects of human "common sense."
Reinforcement Learning (RL)	A type of machine learning where an AI learns by trying actions in an environment and receiving feedback (rewards or penalties) to improve its decisions over time.
Tree Search	A planning algorithm that explores multiple possible sequences of actions (like branches on a tree) to choose the best outcome; used in game-playing AI and robotics planning.

3 Key Takeaways

1. Mo Chen emphasizes that effective robotics requires merging classical AI methods like planning and pathfinding with modern data-driven AI to handle unpredictable, real-world environments.

2. Robots must be designed to understand unspoken social rules, operate safely around people, and know when to pause, ask for help, or defer to human supervision.

3. From hospital delivery robots to "follow-ahead" systems inspired by autonomous luggage, Chen highlights that progress in robotics isn't just technical;it must also include ethical design, transparency, and user trust.

Discussion Questions

1. What kinds of cues should a robot pay attention to so it knows when not to continue a task?

2. Mo Chen notes that in environments where service robots operate, the community has adopted a conservative approach, robots stop when they're uncertain. What are the potential risks and benefits of this strategy, and how might designers create a better balance between safety and efficiency?

3. As robots become more common in workplaces and public spaces, what factors will help people feel comfortable working alongside them?

Relevant Resources

- Mo Chen's Talk at UpperBound, Human Robot Co-Navigation
- Find research publications through the Multi-Agent Robotic Systems Lab
- Canadian Robotics Network

References

Alberta Machine Intelligence Institute. (2025, June 17). How to not get run over by robots with Mo Chen | Approximately Correct Live at UpperBound [Video]. YouTube. https://www.youtube.com/watch?v=1jVSooW2vUc

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